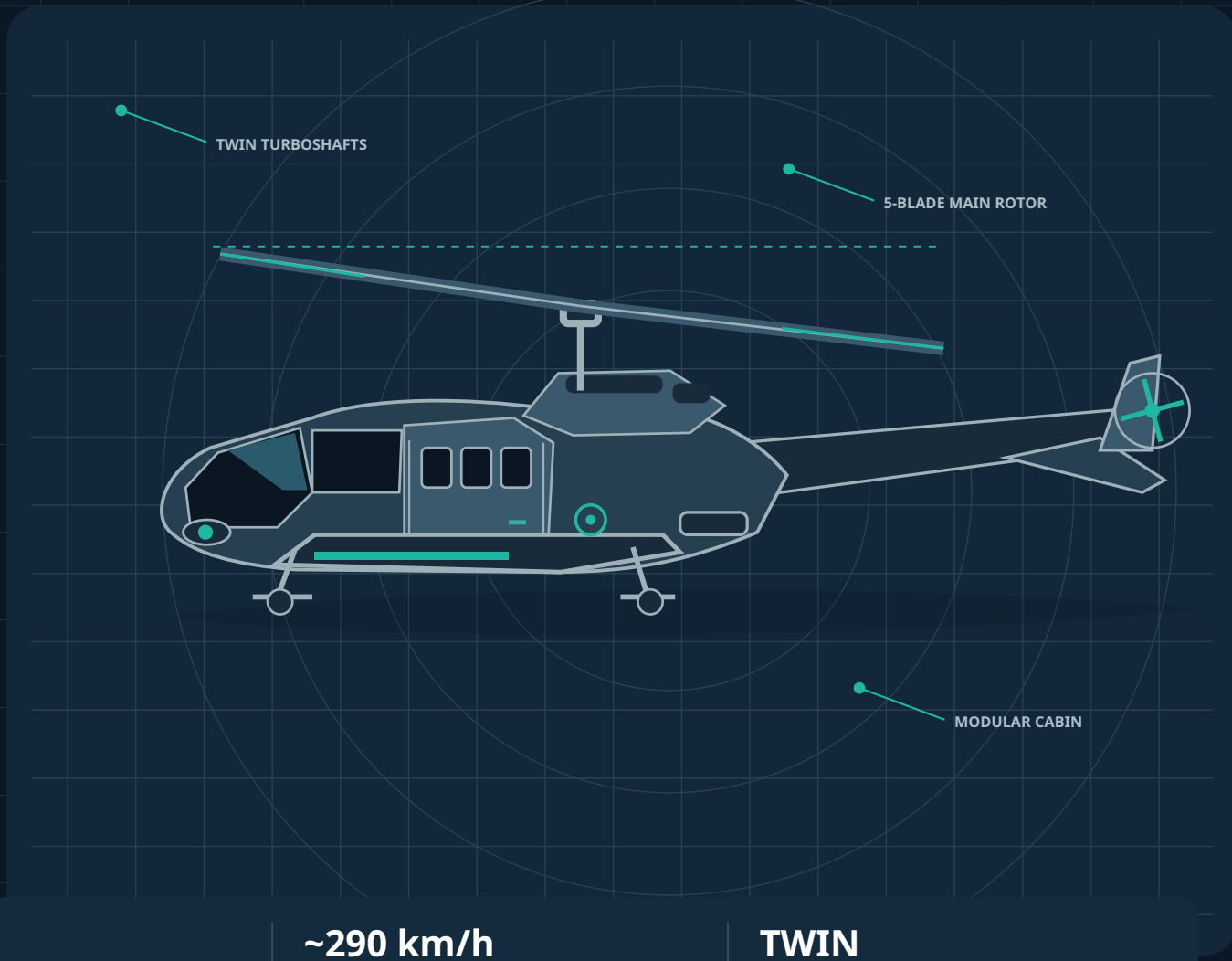
BRITISH ROTORCRAFT CONCEPT / 2026

KESTREL
H-90

NEXT-GENERATION
MULTI-ROLE UTILITY HELICOPTER

A British-led concept for a rugged, connected and adaptable medium-lift aircraft - engineered around mission flexibility, maintainability and future-ready systems.



~10.5 t	12-16	~290 km/h	TWIN
MAX GROSS MASS TARGET	EQUIPPED PERSONNEL*	CRUISE TARGET	ENGINE CONFIGURATION

*Capacity target depends on cabin fit, equipment and certification. All KESTREL figures are unvalidated concept targets, not aircraft specifications.

01 / DESIGN INTENT

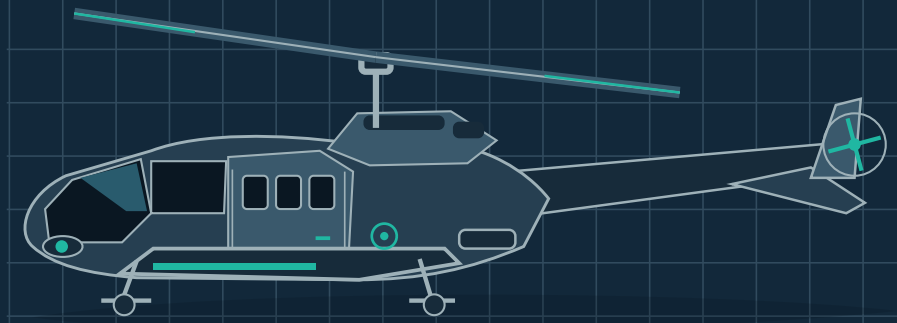
Built around the mission

The KESTREL concept starts with proven utility-helicopter fundamentals, then places digital integration and supportability at the centre of the design.

PROPOSED CONFIGURATION

Conventional rotorcraft architecture

Twin engines, a five-blade main rotor and conventional tail rotor form a credible baseline for development, test and maintainability.



01 | RELIABLE BY DESIGN

A conventional configuration prioritises a clear development path, service access and mature rotorcraft engineering.

02 | DIGITAL AT THE CORE

Open mission architecture, integrated navigation and flight data, and a growth path for assisted-flight functions.

03 | SURVIVABILITY

Crashworthy structures and seats, protected fuel systems and integrated defensive-aids provisions, subject to customer requirements.

04 | LIFE-CYCLE VALUE

Designed for maintainability: accessible modules, built-in health monitoring and simplified support workflows.

CONCEPT DESIGN PRINCIPLE

Make the aircraft straightforward to operate, upgrade and sustain - not merely impressive on a specification sheet.

Design note: the selected baseline is intentionally conventional. Compound/high-speed rotorcraft technology could be explored as a separate future demonstrator, rather than adding unproven complexity to the first aircraft.

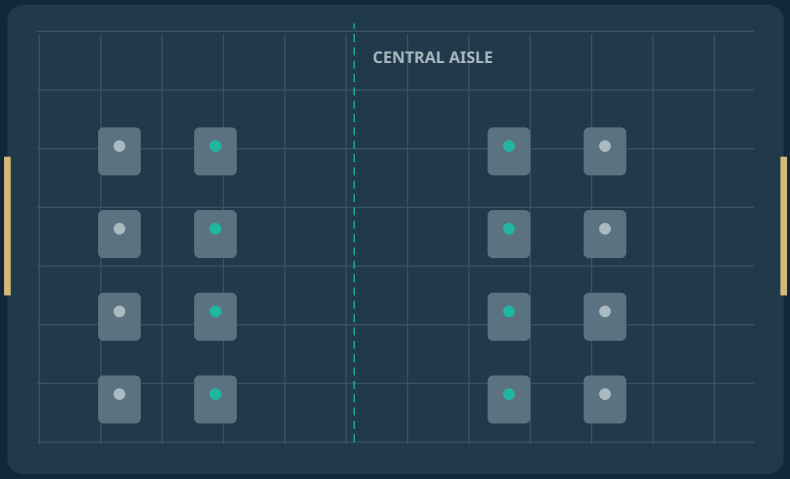
02 / ONE AIRFRAME, MANY ROLES

A configurable mission cabin

A common aircraft family can simplify training, spares and support while adapting cabin equipment to different operators and missions.

CABIN CONFIGURATION STUDY

From people to payload



SLIDING DOORS

Illustrative layout only. Final seating, restraints, floor strength and emergency exits require detailed engineering and certification.

01 / TROOP & CARGO

Removable seating and tie-down arrangements support personnel moves, resupply and internal cargo. External-load capability would be engineered to a defined load envelope.

02 / MEDICAL EVACUATION

Rapid cabin reconfiguration for casualty movement, medical equipment and stretcher layouts; definitive capacity depends on the selected medical fit.

03 / SEARCH & RESCUE

Provision for rescue hoist, searchlight, weather/search sensors and mission console integration, depending on customer variant.

04 / HUMANITARIAN SUPPORT

Disaster relief, remote logistics and evacuation missions supported by cabin access, configurable payloads and field-serviceable equipment.

ONE PLATFORM — Shared training, common spares, variant-specific kits.

03 / DIGITAL, OPEN & SUPPORTABLE

A platform that can evolve

Capability is delivered by the aircraft plus its mission software, sensors, support system and upgrade path - not by airframe performance alone.

FLIGHT & CREW

Digital flight deck

Multi-function displays; integrated navigation and communications; advanced autopilot; crew workload management.

MISSION & PROTECTION

Mission-system integration

Open interfaces for sensors and radios; defensive-aids integration; secure data exchange and mission planning.

HEALTH & SUPPORT

Predictive maintenance

Built-in diagnostics; component health monitoring; modular line-replaceable units; fleet-level maintenance analytics.

SUPPORT PHILOSOPHY

Design for availability across the life of the fleet

01 **MODULARITY**

Replaceable units and clear access

02 **DATA**

Condition and usage monitoring

03 **SUPPLY**

Common spares and configuration control

04 **UPGRADE**

Planned software + hardware growth

Integration principles are proposed objectives. Specific avionics, defensive aids, autonomy functions and data interfaces would be selected with the customer, safety case and export-control requirements.

04 / MARKET POSITION

A credible medium-lift target

KESTREL is conceived to sit in the established twin-engine utility class. Figures below are indicative and are not directly comparable under identical operating conditions.

ATTRIBUTE	SIKORSKY UH-60M	LEONARDO AW149	KESTREL H-90 CONCEPT
Maximum gross mass	9,979 kg	Published performance variants differ	~10,500 kg target
Troop capacity	12 fully equipped, seated troops	Up to 16 heavily equipped / 19 lightly equipped	12-16 equipped personnel target
Cruise speed	268 km/h maximum cruise in S-70 brochure	294 km/h published max cruise at stated conditions	~290 km/h design target
Powerplant	Twin turboshafts	Twin turboshafts	Twin turboshafts
Mission approach	Proven multi-role utility platform	Medium-lift multi-role platform	Modular cabin + open mission architecture

DESIGN TARGETS - TO BE VALIDATED

~10.5 t / ~290 km/h / ~700 km

Initial study targets only. Range definition, payload, reserves, altitude, temperature, rotor performance and fuel assumptions must be set before comparison or customer claims.

WHAT "SIKORSKY-LEVEL" MEANS

Not a styling exercise: prove payload and mission performance; build a reliability and support case; control life-cycle cost; test survivability; and demonstrate a dependable supply chain and upgrade path.

Sources: Lockheed Martin S-70 / UH-60M public material; Leonardo AW149 2024 brochure. Specific variants and test conditions vary; see source notes on final page.

05 / FROM CONCEPT TO AIRCRAFT

A disciplined route to capability

A helicopter of this class requires an experienced rotorcraft prime or development partner, qualified propulsion and transmission suppliers, certification evidence and substantial test investment.

PROPOSED DEVELOPMENT GATES

A — REQUIREMENTS & TRADE STUDY

Mission profiles, payload, range, hot/high, deck/ship needs, customer and export market.

B — INDUSTRIAL PARTNERSHIP

Rotorcraft OEM, engine, gearbox, rotor, flight controls, structures and certification partners.

C — ENGINEERING & DEMONSTRATORS

Digital mock-up, drivetrain rigs, rotor test, systems integration, maintainability and safety cases.

D — PROTOTYPE, TEST & CERTIFY

Ground runs, envelope expansion, flight test, reliability growth and customer qualification.

WHERE BAE COULD ADD VALUE

Systems-led integration

A differentiated role could centre on mission systems, flight-control technology, defensive aids, secure integration, autonomy interfaces and fleet support - alongside an established rotorcraft manufacturer.

MARKET REALITY

The UK's 2026 New Medium Helicopter contract selected the Leonardo AW149. KESTREL should therefore be assessed as a future/export concept, not as the selected NMH platform.

STATUS

Independent concept proposal for discussion. No official BAE Systems programme, confirmed supplier partnership, prototype or validated performance data is implied.

PUBLIC REFERENCE POINTS

1 UK Ministry of Defence / GOV.UK, "Billion-pound helicopter deal secures 3,300 British
2 Lockheed Martin, "UH-60M Black Hawk Helicopter" product information
3 Lockheed Martin / Sikorsky, "S-70 Multi-Mission Helicopter" brochure (published perfor
4 Leonardo Helicopters, "AW149 brochure" (2024)
5 BAE Systems, Air / flight-control systems and defensive-aids capabilities